CWT CO2 sensor (RS485 type) manual

Product description

The transmitter is widely used in agricultural greenhouses, flower cultivation and other occasions that require CO2, illuminance, temperature and humidity monitoring. The input power supply, the sensor probe and the signal output in the sensor are completely isolated. Safe and reliable, beautiful appearance, easy installation.

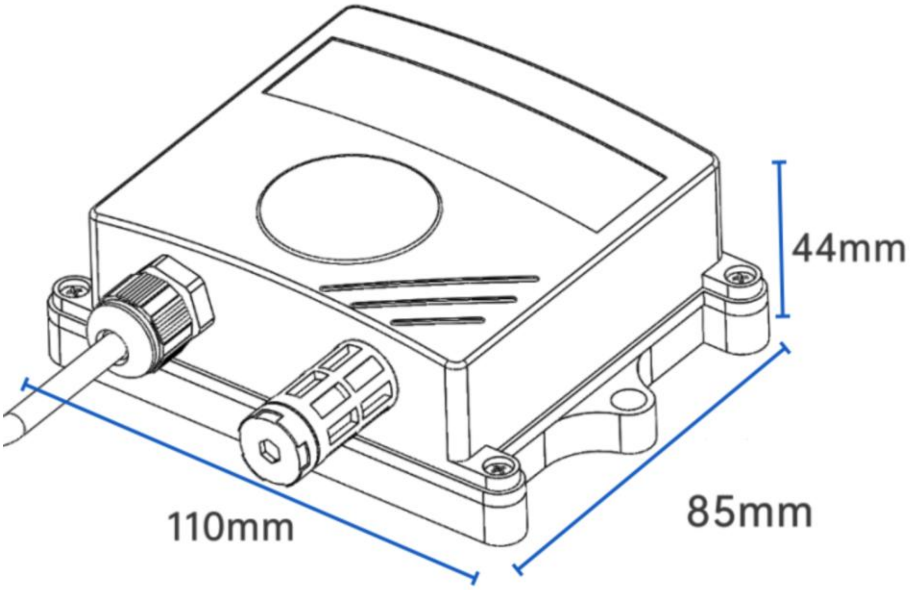
Features This product uses a high-sensitivity gas detection probe with stable signal and high accuracy. It has the characteristics of wide measurement range, good linearity, convenient use, easy installation, and long transmission distance. It is suitable for indoor and outdoor use. The IPV65 shell is fully waterproof and can be used in various harsh environments.

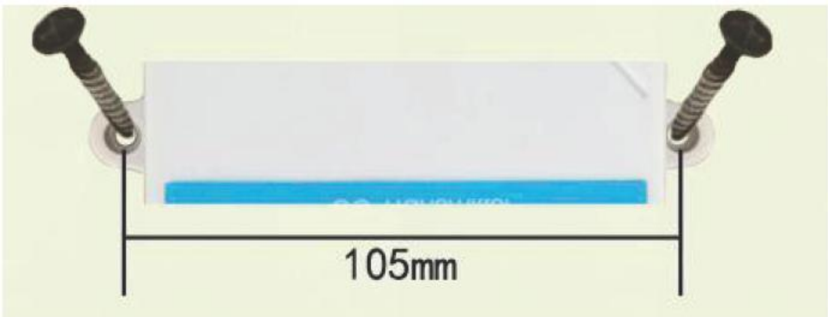
Parameters measuring

Item	Details
Power Consumption	0.3W (24VDC)
Power Supply	10~30V DC (Average Current < 85mA)
CO_2 Measuring Range	0~5000ppm (Effective Range: 400-5000) Optional: 0~2000ppm (Effective Range: 400-2000) 0~10000ppm (Effective Range: 400-10000)
CO_2 Accuracy	0~5000ppm: ± (50ppm + 3% F•S) (25℃) 0~10000ppm: ± (50ppm + 5% F•S) (25℃) High Precision: 0~5000ppm: ± (45ppm + 3% F•S) (25℃) 0~10000ppm: ± (45ppm + 5% F•S) (25℃)
System Warm-up Time	2min (Usable), 10min (Maximum Precision)
Response Time	Typically < 180s for 90% Step Change
Stability	< 2% F•S
Non-linearity	< 1% F•S
Resolution	1ppm
Temperature Measuring Range	-40℃~+80℃
Temperature Accuracy	±0.5℃ (25℃)
Temperature Resolution	0.1℃

Humidity Measuring Range	0~100%RH
Humidity Accuracy	±3%RH (60%RH, 25℃)
Humidity Resolution	0.1%RH
Operating Environment	-20℃~+60℃, 0%RH~95%RH (Non-condensing)
Data Update Interval	2s

Size





Wiring

Cable color	description
Brown	Power + (DC10-30V)
black	Power -
yellow	RS485 A+
blue	RS485 B-

RS485 communication

Default parameters: 4800,n,8,1
Default device address is 1
Modbus RTU protocol

Single CO2 Sensor

Read status registers, read function code: 0x03/04							
Register address (Hex)	PLC Address (decimal)	meaning	Number of bytes	Data type	Scale	Range	R/W
0000	40001	CO2	2	Uint16	1	0-2000/5000/10000/20000/50000ppm	read
0002	40003	CO2	2	Uint16	1		
0000	40001	CO2 high 16 byte	2	Uint32	1	0-100000ppm	read
0001	40002	CO2 low 16 byte	2		1		read

CO2, Temperature and Humidity Integrated Sensor

Read status registers, read function code: 0x03/04							
Register address (Hex)	PLC Address (decimal)	meaning	Number of bytes	Data type	Scale	Range	R/W
0000	40001	Humidity	2	Uint16	0.1	0~1000	R
0001	40002	Temperature	2	Int16	0.1	-400~1000	R
0002	40003	CO2	2	Uint16	1	0-2000/5000/10000/ 20000/50000ppm	R
0002	40003	CO2 high 16 byte	2	Uint32	1	0-100000ppm	R
0003	40004	CO2 low 16 byte	2				
0032	40051	Temperature offset	2	Int16	0.1	-400~1000	R/W
0035	40054	Humidity offset	2	Int16	0.1	-400~1000	R/W
0038	40057	CO2 offset	2	Int16	1	-2000~2000	R/W
Parameters registers, read function code: 0x03/04, write function code: 0x06							
07D0	42001	Slave ID	2			1-254	R/W
07D1	42002	baud rate	2			0: 2400 1: 4800(default) 2: 9600 3: 19200 4: 38400 5: 57600 6: 115200 7: 1200	R/W

E.g., read CO2:
Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x00	0x02	0x00	0x01	0x25	0xCA

Sensor responds:

Address	Function Code	Number of byte	CO2 value	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x02	0x0B 0xB8	0xBF	0x06

CO2: BB8 H= 3000 => CO2 = 3000ppm

E.g., read Humidity, temperature, CO2 together:
Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x00	0x00	0x00	0x03	0x05	0xCB

Sensor responds:

Address	Function Code	Number of byte	humidity value	temperature value	CO2 value	Error Check (Lo)	Error Check (Hi)
0x01	0x03	0x06	0x01 0x67	0xFF 0xB5	0x0B 0xB8	0x33	0xDC

Temperature calculates:
When temperature less than 0, value will be responded in complement
Temperature: FF85 H= -75=> temperature= -7.5℃
Humidity: 167 H= 359 => humidity= 35.9%
CO2: BB8 H= 3000 => CO2 = 3000ppm

Set slave ID

E.g., set slave ID=2, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD0	0x00 0x02	0x08	0x86

Sensor responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	ID	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD0	0x00 0x02	0x08	0x86

Set baud rate

E.g., set baud rate to 9600, Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD1	0x00 0x02	0x59	0x46

Sensor responds:

Address	Function Code	Start Address (Hi)	Start Address (Lo)	command	Error Check (Lo)	Error Check (Hi)
0x01	0x06	0x07	0xD1	0x00 0x02	0x59	0x46

Enquiry slave ID

Master sends

Address	Function Code	Start Address (Hi)	Start Address (Lo)	Number of Points (Hi)	Number of Points (Lo)	Error Check (Lo)	Error Check (Hi)
0xFF	0x03	0x07	0xD0	0x00	0x01	0x91	0x59

Sensor responds:

Address	Function Code	Number of Points	address	Error Check (Lo)	Error Check (Hi)
0xFF	0x03	0x02	0x00 0x01	0x50	0x50